

Gr. 11 March Test

Exponents Memo

$$\begin{aligned}
 1.1. \quad & \frac{\sqrt{32}}{2} + \sqrt{2} \\
 &= \frac{4\sqrt{2}}{2} + \sqrt{2} \\
 &= 2\sqrt{2} + \sqrt{2} \\
 &= 3\sqrt{2}
 \end{aligned}$$

$$\begin{aligned}
 1.2 \quad & \sqrt[3]{8} + \sqrt[3]{27} + \sqrt[3]{-64} \\
 &= 2 + 3 - 4 \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 1.3. \quad & \frac{6}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} \\
 &= \frac{6\sqrt{3}}{3} \\
 &= 2\sqrt{3}
 \end{aligned}$$

$$\begin{aligned}
 1.4 \quad & \left[\frac{p^{\frac{3}{4}}}{r^{\frac{2}{3}}} \right] \cdot \sqrt[3]{p^{-1}} \cdot p^{\frac{1}{4}} \\
 &= \frac{p^{\frac{3}{4}}}{r^{\frac{2}{3}}} \cdot p^{-\frac{1}{3}} \cdot p^{\frac{1}{4}} \\
 &= \frac{p^{\frac{3}{4} + \frac{1}{4}}}{r^{\frac{2}{3} + \frac{1}{3}}} \\
 &= \frac{p}{r}
 \end{aligned}$$

$$\begin{aligned}
 1.5 \quad & \frac{2^{9+4} - 6 \cdot 2^{9+1}}{5 \cdot 2^9 \cdot 2^2} \\
 &= \frac{2^9(2^4 - 6 \cdot 2)}{5 \cdot 2^9 \cdot 2^2} \\
 &= \frac{16 - 12}{5 \cdot 4} \\
 &= \frac{1}{5}
 \end{aligned}$$

$$\begin{aligned}
 1.6 \quad & \frac{4^{n+2} \cdot 9^{n-1}}{72^n \cdot 2^{1-n}} \\
 &= \frac{2^{2n+4} \cdot 3^{2n-2}}{2^{3n} \cdot 3^{2n} \cdot 2^{1-n}} \\
 &= 2^{2n+4-3n-1+n} \cdot 3^{2n-2-2n} \\
 &= 2^3 \cdot 3^{-2} \\
 &= \frac{8}{9}
 \end{aligned}$$

$$\begin{aligned}
 2.1 \quad & 3 \cdot 2^x = 48 \\
 & 2^x = 16 \\
 & x = 4
 \end{aligned}$$

$$\begin{aligned}
 2.2 \quad & 9^x + 3^{2x+1} = 108 \\
 & 3^{2x} + 3^{2x+1} = 108 \\
 & 3^{2x}(1+3) = 108 \\
 & 3^{2x} = 27 \\
 & \therefore 2x = 3 \\
 & x = \frac{3}{2}
 \end{aligned}$$

$$\begin{aligned}
 3. \quad & \frac{2^{2017} + 2^{2019}}{2^{2018}} \\
 &= \frac{2^{2018}(2^{-1} + 2^1)}{2^{2018}} \\
 &= \frac{1}{2} + 2 \\
 &= \frac{5}{2}
 \end{aligned}$$

$$2.3 \quad 2^{2x} + 2^x - 6 = 0$$

$$(2^x + 3)(2^x - 2) = 0 \quad \checkmark \checkmark$$

$$2^x = -3$$

$$2^x = 2$$

$$N/A \quad \checkmark$$

$$x = 1 \quad \checkmark$$

4

$$2.4. \quad 3(x-1)^{\frac{1}{2}} = 12$$

$$(x-1)^{\frac{1}{2}} = 4 \quad \checkmark$$

$$x-1 = 16 \quad \checkmark$$

$$x = 17 \quad \checkmark$$

3

7

35